AGU 397 SUMMER INTERNSHIP REPORT

SUBMITTED TO

THE DEPARTMENT OF INDUSTRIAL ENGINEERING OF

ABDULLAH GUL UNIVERSITY

IN PARTIAL FULLFILLMENT OF THE REQUIREMENTS

FOR THE DEGREE OF BACHELOR OF SCIENCE

Menarini Sağlık ve İlaç Sanayi Ticaret A.Ş.

01.07.2025-12.08.2025

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<20.10.2025>

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SUMMARY

This report summarizes the internship experience I gained during my time at Menarini Group. My project focused on improving the production planning process by developing a Master Production Schedule (MPS) and Material Requirements Planning (MRP) system based on forecasted demand. I utilized various forecasting methods and applied them to historical sales data to build a more accurate planning model. I had the opportunity to work with real-life constraints such as shutdown periods, safety stock levels, and lead times.

Throughout the internship, I am not only performed demand forecasting and production scheduling in Excel, but also developed a Python based interface to automate MPS and MRP calculations. This experience allowed me to strengthen my skills in data analysis, programming, and industrial engineering principles. Additionally, I gained insight into the operational structure of Menarini and how digital tools can enhance decision making processes.

TABLE OF CONTENTS

1.INTRODUCTION..... 1
2. THE SYSTEM UNDER CONSIDERATION..... 3
2.1 Overview of the Production Planning Process.....3
2.2 Coordination Between Departments.....4
2.3 Organizational Structure of the Company.....4
2.4 Production Scheduling Workflow..... 5
3.METHADOLOGY..... 6
3.1 Problem Analysis.....6
3.2 Literature Review..... 11
3.3. Model Building Development of a Python Based Interface..... 12
5.Conclusion..... 14
6.Bibliography..... 15
Appendices..... 15

LIST OF FIGURES

Figure 1 Production Facility of Menarini Turkey 2
Figure 2 One of the most popular product of Menarini 2
Figure 3 Process Dıagram 3
Figure 4 Coordination between planning department and others 4
Figure 5 Organizational Structure of the Company 4
Figure 6 Company Organization Chart 4
Figure 7 Process Scheduling Workflow Dıagram 5
Figure 8 Holt Winter Forecast Graph 7
Figure 9 Master Production Schedule of Product B 7
Figure 10 BOM Chain 8
Figure 11 Material Requirements Planning of Sorbitol 8
Figure 12 Integrated Excel Model for MPS and MRP Planning 8
Figure 13 Demand,Production and Stock Quantity of Poduct A 9
Figure 14 Demand, Production and Stock Quantity of Product B 9
Figure 15 Monthly Raw Material Status of Sorbitol 10
Figure 16 Monthly Raw Material Status of Karbomer 11
Figure 18 MRP and MPS Interface 12
Figure 17 MRP and MPS Interface 12
Figure 19 MPS and MRP Overview Including Forecast, Production, and Raw Material Requirements 13

1.INTRODUCTION

The internship was carried out at Menarini Turkey Health and Pharmaceutical Industry Trade Inc., which operates in the pharmaceutical industry. The company is located in Zeytinburnu, Istanbul, and was established in 1955. The ownership of the company currently belongs to Uğur Bingöl.

Menarini Turkey operates under the international pharmaceutical group Menarini, an Italian company founded in Naples in 1886 under the name “Farmacia Internazionale”. In 1915, the company moved its headquarters to Florence, which remains its main operational hub today. With over 135 years of history, Menarini Group bases its operations on five core principles: providing high-quality therapeutic and diagnostic solutions for patients, maintaining ethics as a fundamental principle, fostering innovation and progress, being strongly human-oriented, and ensuring environmental sustainability [1].

Menarini Group operates in 140 countries worldwide, spanning Europe, Asia, Africa, the Middle East, and Central America. Recently, the Group entered the U.S. oncology market by acquiring Stemline Therapeutics, a biopharmaceutical company listed on NASDAQ [1].

With more than 17,000 employees, the Group operates 18 production facilities, including a biotechnology plant dedicated to the production of monoclonal antibodies, manufacturing over 500 million drugs annually. These drugs are distributed across six continents, primarily focusing on cardiology, oncology, pneumology, inflammation, and gastroenterology [1].

The company’s total production area in Turkey includes 22,000 m² of outdoor space and 10,300 m² of indoor facilities. Menarini Turkey employs approximately 700 people.

Menarini Turkey has three production units that are beta-lactam group products, oral solid and small volume sterile forms, topical forms. Beta lactam group products are like penicillin and cephalosporin-based antibiotics. Oral solid and small wolume sterile forms such as ampoules and drops. Additionally, topical forms like creams , gels and pomades.

Products such as ARVELES , Bilaxen, and Nidazole syrup are among the well-known pharmaceuticals manufactured at the facility. The company achieves an annual production capacity of 100 million boxes, supplying both the domestic and international markets. Raw materials are sourced from various countries, including China and India.

Menarini Turkey uses the GMP (Good Manufacturing Practice) standard, which ensures high-quality and hygienic production conditions by controlling environmental factors throughout all stages from raw material input to distribution. The company primarily serves pharmacies and hospitals, including specialized production for hospitals and contract manufacturing.

Figure 1 Production Facility of Menarini Turkey

*Figure SEQ Figure * ARABIC 2 One of the most popular product of Menarini*

2. THE SYSTEM UNDER CONSIDERATION

During my internship at the Production Planning Department of Menarini Turkey, I had the opportunity to observe and analyze the production planning and coordination processes in detail. This section explains the current status of the system, supported by process diagrams, organizational structure, and interdepartmental workflows.

2.1 Overview of the Production Planning Process

The production planning process begins with the reception of sales forecasts, which are prepared in three main periods: Budget, Forecast, and Reforecast. The Budget forecast is created in August–September for the following year. The Forecast is updated in March–April for the remaining months of the current year, and the Reforecast is revised again in August–September based on updated market conditions.

Once the most recent forecast is received, a two month stock check is performed. If stock is sufficient, production is scheduled; if not, it is postponed. Material availability is then verified. In case of shortages, the Purchasing Department is consulted, and the plan is revised according to material lead times. This cyclical and forecast driven approach provide efficient resource use and synchronization between demand, inventory, and procurement.

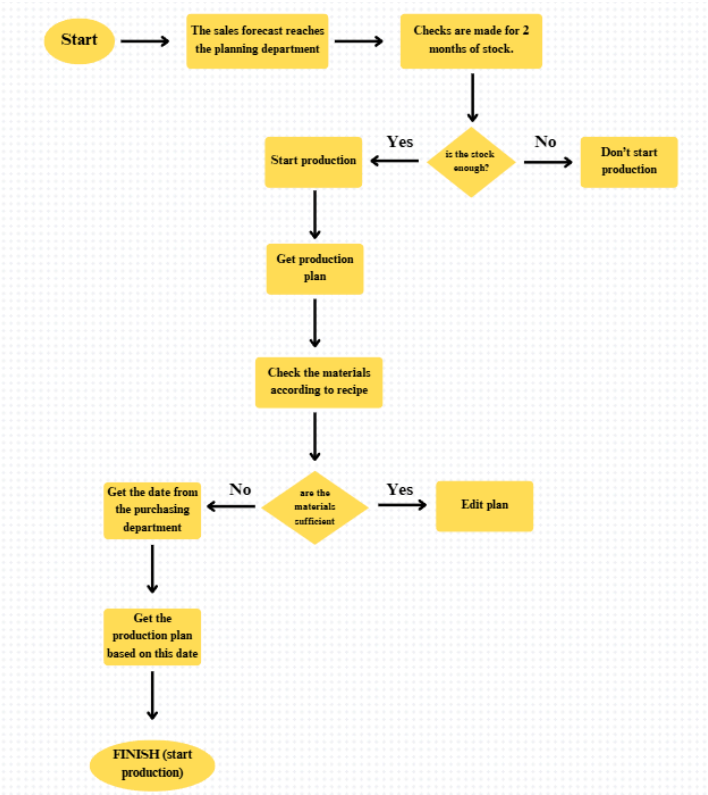

Figure SEQ Figure * ARABIC 3 Process Diagram

2.2 Coordination Between Departments

The Planning Department serves as the central link between multiple units. The Sales and Marketing Department provides forecasts which are used by Planning to prepare schedules. The Purchasing Department is then responsible for sourcing the required materials. Once everything is aligned, the production department carries out the actual production.

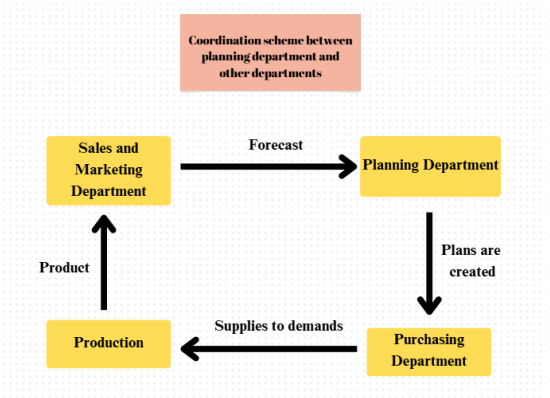

*Figure SEQ Figure * ARABIC 4 Coordination between planning department and others*

2.3 Organizational Structure of the Company

The Planning Department operates under the Factory Director, along with other departments such as Quality Control, Quality Assurance, Engineering, and Production. The top-level hierarchy includes the META Regional President and the General Directorate of Turkey, followed by various departmental directors.

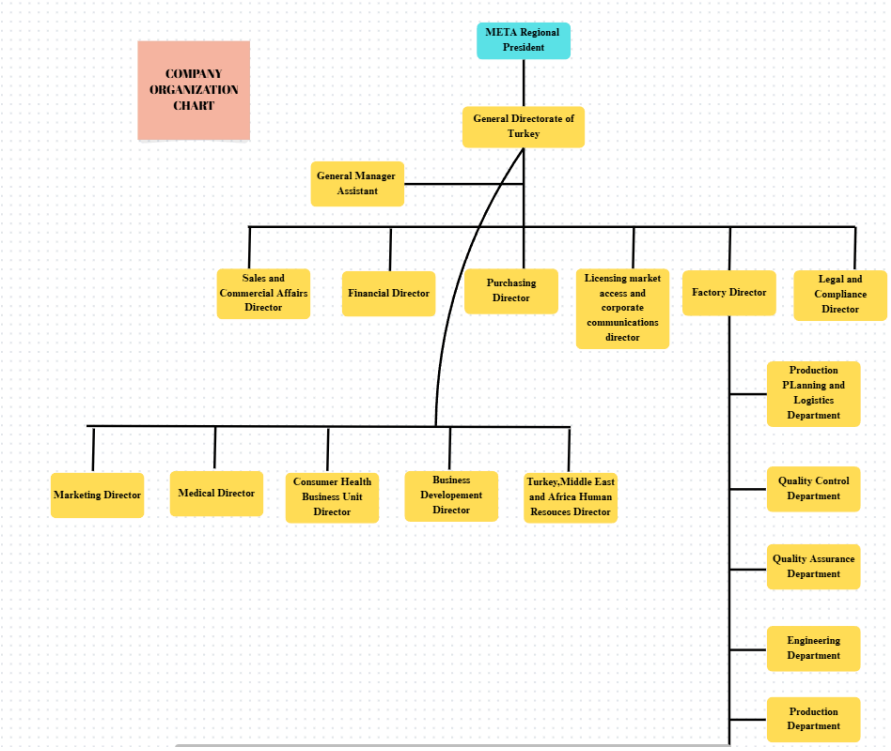

*Figure SEQ Figure * ARABIC 6 Company Organization Chart*

2.4 Production Scheduling Workflow

Another key responsibility of the Planning Department is production scheduling. After receiving orders from sales or customers, demand and resource availability are assessed. A production plan is created, materials are procured and coordination with the Production Department is maintained. The process is monitored and if any issues arise, the schedule is adjusted; otherwise, production continues until delivery is ready.

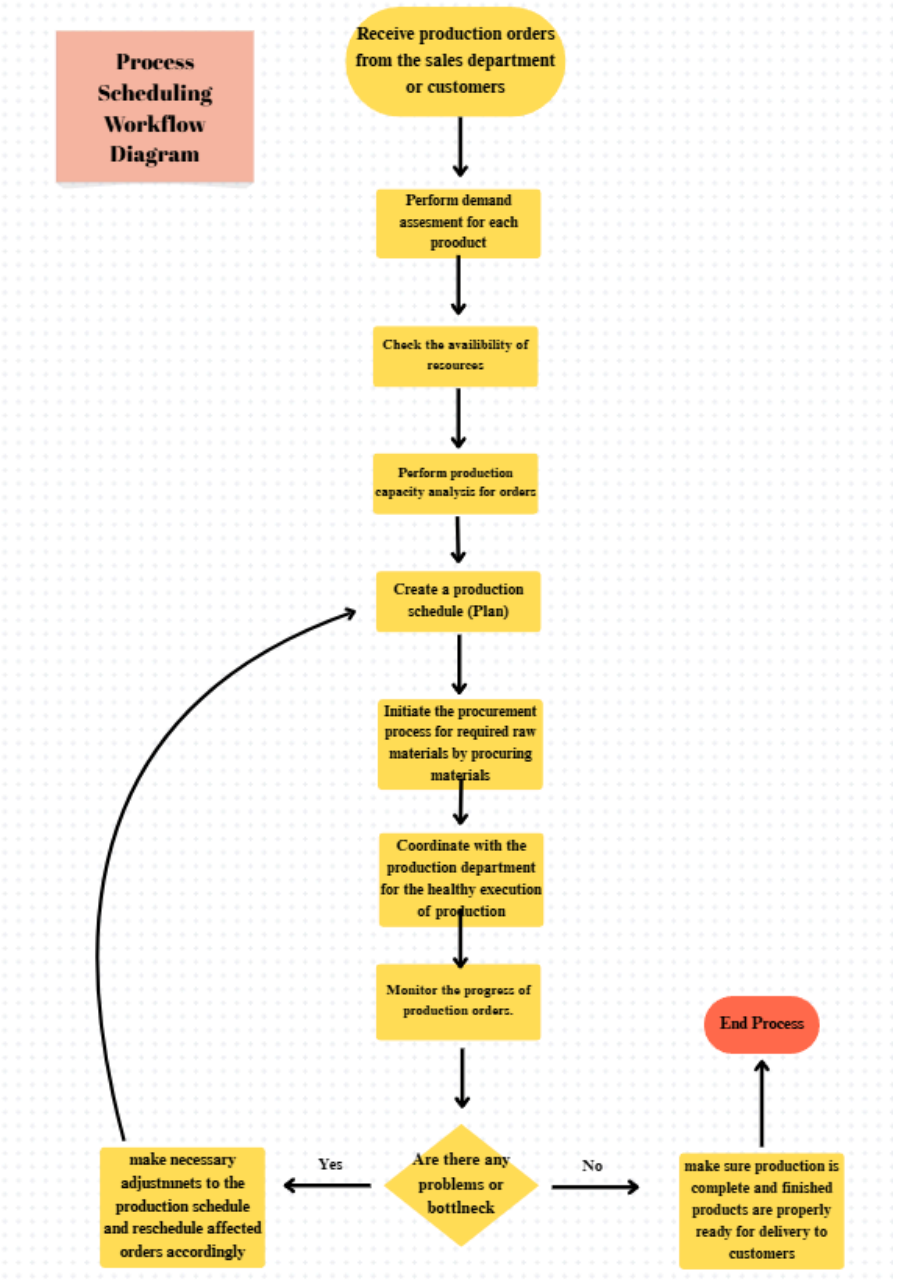

*Figure SEQ Figure * ARABIC 7 Process Scheduling Workflow Diagram*

3.METHADOLOGY

During my internship at Menarini Turkey's Planning Department, I observed several inefficiencies in the current production planning workflow, particularly in how Master Production Scheduling (MPS) and Material Requirements Planning (MRP) are calculated and managed.

In the current system, MPS and MRP calculations are performed manually using separate Excel spreadsheets. Although the necessary data is extracted from the SAP system, the planning calculations are not integrated into SAP itself. Instead, planners export raw data from SAP and then process it independently in Excel. This divided workflow creates three major problems. Performing calculations outside the ERP system using separate Excel files leads to manual errors, inconsistent results, and a lack of integration between MPS and MRP. This disconnect prevents real-time, end-to-end visibility of stock and material needs, slows down decision-making, and heavily relies on the planner’s personal judgment. As a result, the risk of overstocking or shortages increases, negatively impacting production efficiency and costs.

3.1 Problem Analysis

The project started with 24 months of historical sales data and several planning constraints, including a mandatory two-month safety stock policy, an initial inventory level, and a production shutdown scheduled for April and lead time . The first step involved generating 12-month demand forecasts for two different products, referred to as Product A and Product B. To achieve this, I applied various time series forecasting methods such as Moving Average (MA), Weighted Moving Average (WMA), Exponential Smoothing, Holt’s Linear Trend and Holt-Winters Seasonal Method.After conducting error analysis using metrics like MAPE and MAE, I determined that the Holt-Winters method provided the most accurate results and therefore used its output as the basis for the next phases of the project. This decision was also supported by the fact that both Product A and Product B exhibit strong seasonal demand patterns. Their sales tend to increase or decrease depending on the time of year, which made it essential to use a forecasting model that can effectively account for trend and seasonality.

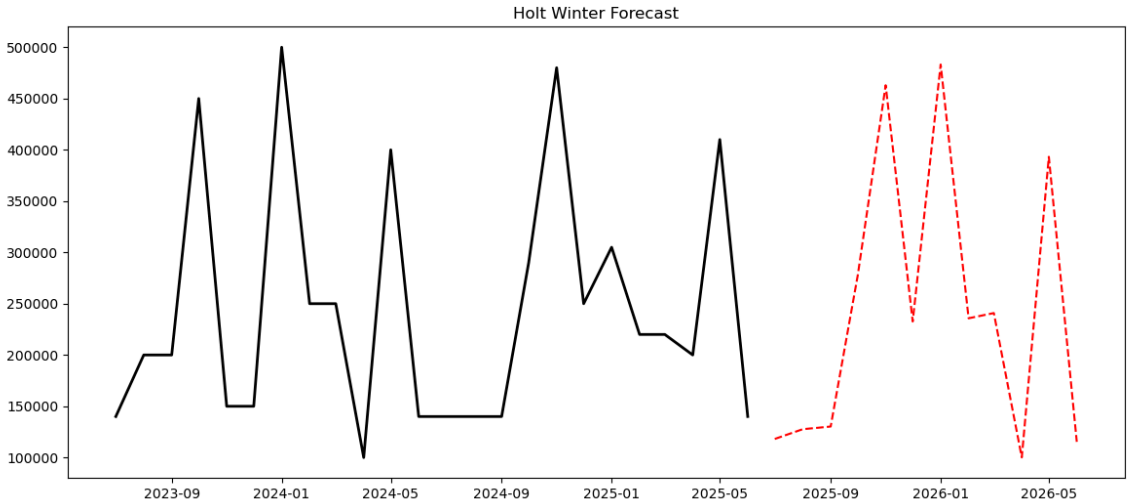

Figure 8 Holt Winter Forecast Graph

Following the forecasting stage, I developed the Master Production Schedule (MPS) for both products using Excel. The MPS was designed to satisfy forecasted demand while complying with the shutdown constraint in April and maintaining the required inventory levels. Based on this schedule, I then performed Material Requirements Planning (MRP) to determine raw material needs. One of the challenges in this step was the existence of shared raw materials between Product A and Product B, which required careful coordination in demand aggregation and calculation of net requirements. The MRP calculations took into account current raw material inventories, lead times and production quantities, ensuring a consistent and optimized procurement plan.

MPS-B

Months	Tem.25	Ağu.25	Eyl.25	Eki.25	Kas.25	Ara.25	Oca.26	Şub.26	Mar.26	Nis.26	May.26
Demand forecast	120000	110000	130000	150000	300000	320000	350000	330000	180000	160000	140000
Starting Stock	180000	310000	450000	595000	720000	695000	650000	550000	420000	440000	280000
Production Quantity	250000	250000	275000	275000	275000	275000	250000	200000	200000	0	100000
Stock on hand	60000	200000	320000	445000	420000	375000	300000	220000	240000	280000	140000
Ending Stock	310000	450000	595000	720000	695000	650000	550000	420000	440000	280000	240000
Stock Level (Month)	2,583333333	3,214285714	2,644444444	2,322580645	2,074626866	1,911764706	2,156862745	2,470588235	2,933333333	2,074074074	1,846153846

Figure 9 Master Production Schedule of Product B

Bill of Materials (BOM)					
A			B		
Lead time	Rate	Raw Material	Raw Material	Rate	Lead time
1	0,5	Ethyl Alcol	Vazelin Solid	0,25	3
2	0,3	Sorbitol	Sorbitol	0,3	2
2	0,1	Lavanta Esansı	Lavanta Esansı	0,1	2
1	0,05	Karbomer	Karbomer	0,05	1
3	0,2	Metil Sodyum	Etil Sodyum	0,12	1

*Figure SEQ Figure * ARABIC 10 BOM Chain*

Sorbitol	0,3					
Months	Production Quantity	Needed	Starting Stock	Stock on hand	Order Quantity	Lead Time
Tem.25	610000	183000	500000	317000	400000	2
Ağu.25	550000	165000	317000	152000		2
Eyl.25	615000	184500	552000	367500	400000	2
Eki.25	635000	190500	367500	177000		2
Kas.25	635000	190500	577000	386500		2
Ara.25	515000	154500	386500	232000	400000	2
Oca.26	510000	153000	232000	79000		2
Şub.26	480000	144000	479000	335000		2
Mar.26	480000	144000	335000	191000		2
Nis.26	0	0	191000	191000	100000	2
May.26	360000	108000	191000	83000		2
Haz.26	75000	22500	183000	160500		2

*Figure SEQ Figure * ARABIC 11 Material Requirements Planning of Sorbitol*

After completing both the MPS and MRP stages, I combined them into a single Excel model, which also included automated graphs to visualize production quantities, raw material needs, and inventory levels over the planning horizon.

Type of Produc	Months	Demand Forecast	Starting Stock	Production Quantity	Stock on hand	Ending Stock	Type of RM	Needed RM	Starting Stock RM	Stock on hand RM	Order Quantity RM	LT
A	Tem.25	118155	125000	360000	6845	366845	Ethyl Alcohol	180000	250000	70000	300000	1
A	Tem.25	118155	125000	360000	6845	366845	Sorbitol	108000	400000	292000	0	2
A	Tem.25	118155	125000	360000	6845	366845	Lavender Essence	36000	120000	84000	80000	2
A	Tem.25	118155	125000	360000	6845	366845	Karbomer	18000	60000	42000	60000	1
A	Tem.25	118155	125000	360000	6845	366845	Metil Sodyum	72000	140000	68000	100000	3
A	Ağu.25	127639	366845	300000	239206	539206	Ethyl Alcohol	150000	370000	220000	300000	1
A	Ağu.25	127639	366845	300000	239206	539206	Sorbitol	90000	292000	202000	260000	2
A	Ağu.25	127639	366845	300000	239206	539206	Lavender Essence	30000	84000	54000	0	2
A	Ağu.25	127639	366845	300000	239206	539206	Karbomer	15000	102000	87000	0	1
A	Ağu.25	127639	366845	300000	239206	539206	Metil Sodyum	30000	68000	38000	80000	3
A	Eyl.25	130243	539206	340000	408963	748963	Ethyl Alcohol	170000	520000	350000	300000	1
A	Eyl.25	130243	539206	340000	408963	748963	Sorbitol	102000	202000	100000	0	2
A	Eyl.25	130243	539206	340000	408963	748963	Lavender Essence	34000	134000	100000	80000	2
A	Eyl.25	130243	539206	340000	408963	748963	Karbomer	17000	87000	70000	60000	1
A	Eyl.25	130243	539206	340000	408963	748963	Metil Sodyum	34000	38000	4000	0	3
A	Eki.25	277441	748963	360000	471522	831522	Ethyl Alcohol	180000	650000	470000	0	1
A	Eki.25	277441	748963	360000	471522	831522	Sorbitol	108000	360000	252000	0	2
A	Eki.25	277441	748963	360000	471522	831522	Lavender Essence	36000	100000	64000	0	2
A	Eki.25	277441	748963	360000	471522	831522	Karbomer	18000	130000	112000	0	1
A	Eki.25	277441	748963	360000	471522	831522	Metil Sodyum	36000	104000	68000	0	3
A	Kas.25	462915	831522	360000	368607	728607	Ethyl Alcohol	180000	470000	290000	0	1
A	Kas.25	462915	831522	360000	368607	728607	Sorbitol	108000	252000	144000	260000	2
A	Kas.25	462915	831522	360000	368607	728607	Lavender Essence	36000	144000	108000	0	2
A	Kas.25	462915	831522	360000	368607	728607	Karbomer	18000	112000	94000	0	1
A	Kas.25	462915	831522	360000	368607	728607	Metil Sodyum	36000	148000	112000	0	3
A	Ara.25	232626	728607	240000	495981	735981	Ethyl Alcohol	120000	290000	170000	300000	1
A	Ara.25	232626	728607	240000	495981	735981	Sorbitol	72000	144000	72000	0	2
A	Ara.25	232626	728607	240000	495981	735981	Lavender Essence	24000	188000	164000	0	2
A	Ara.25	232626	728607	240000	495981	735981	Karbomer	12000	94000	82000	0	1
A	Ara.25	232626	728607	240000	495981	735981	Metil Sodyum	24000	112000	88000	0	3
A	Oca.26	483138	735981	260000	252843	512843	Ethyl Alcohol	130000	470000	340000	0	1
A	Oca.26	483138	735981	260000	252843	512843	Sorbitol	78000	332000	254000	0	2
A	Oca.26	483138	735981	260000	252843	512843	Lavender Essence	26000	84000	58000	80000	2
A	Oca.26	483138	735981	260000	252843	512843	Karbomer	13000	82000	69000	0	1
A	Oca.26	483138	735981	260000	252843	512843	Metil Sodyum	26000	88000	62000	0	3
A	Şub.26	235759	512843	280000	277084	557084	Ethyl Alcohol	140000	340000	200000	0	1
A	Şub.26	235759	512843	280000	277084	557084	Sorbitol	84000	254000	170000	0	2
A	Şub.26	235759	512843	280000	277084	557084	Lavender Essence	28000	58000	30000	0	2
A	Şub.26	235759	512843	280000	277084	557084	Karbomer	14000	69000	55000	0	1

Figure 12 Integrated Excel Model for MPS and MRP Planning

Line Plot

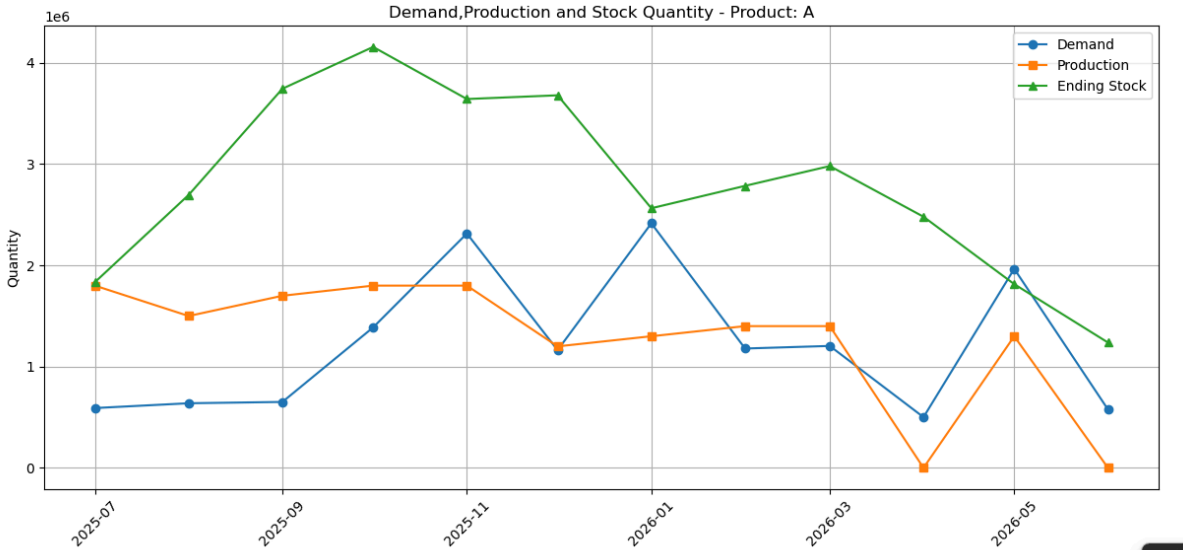

Figure 13 Demand,Production and Stock Quantity of Poduct A

The graph shows that production quantities are consistently above or close to demand, which leads to a steady increase in ending stock. This stock build up appears intentional and strategic, preparing for the planned production shutdown in April, where production drops to zero. Despite no production in April, the accumulated stock is sufficient to meet demand, indicating that the safety stock policy and forward planning were effective in maintaining service levels during the shutdown period. Following the planned shutdown in April, facility works continue for a total of three weeks in May, with the first week allocated to cleaning. This results in a reduction in production during this period

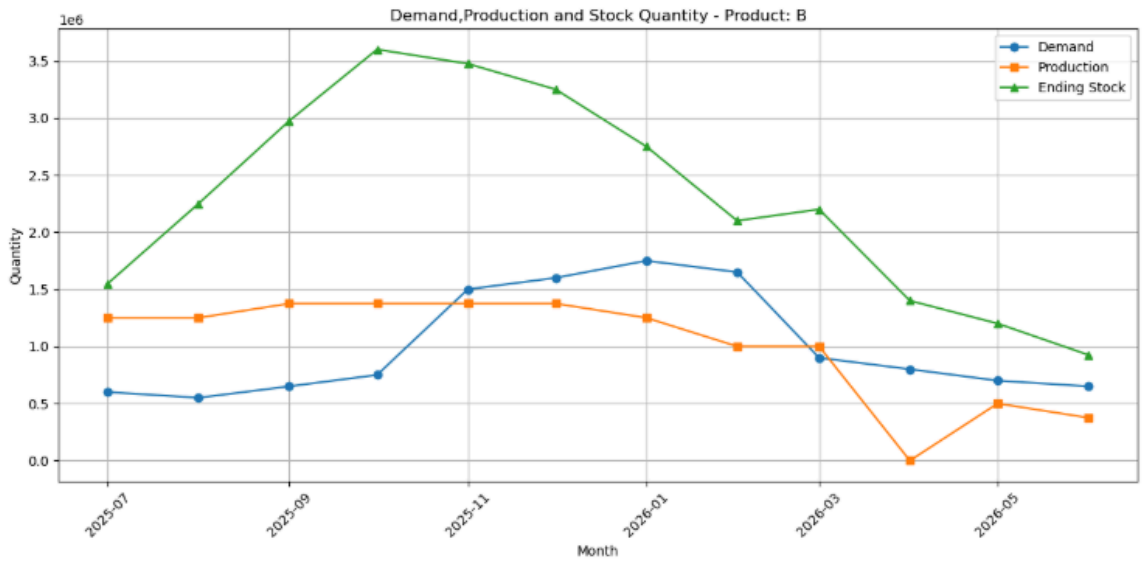

*Figure SEQ Figure * ARABIC 14 Demand, Production and Stock Quantity of Product B*

In line with the strategy of maintaining a 2 month safety stock, production was kept above demand in the early months, leading to stock accumulation. This buffer stock was intentionally built up to cover the April production shutdown. As a result, even with zero production during April, customer demand was met without disruption, and the service level was successfully maintained.

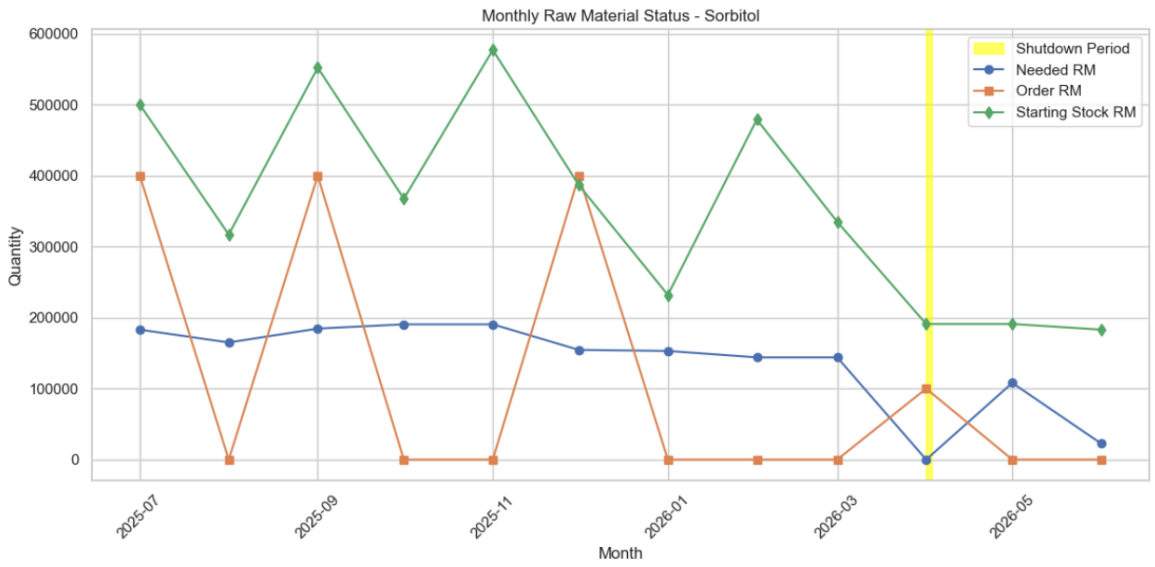

*Figure SEQ Figure * ARABIC 15 Monthly Raw Material Status of Sorbitol*

Sorbitol is a shared raw material used in the production of both Product A and Product B. Therefore, its requirement was calculated based on the combined production quantities of both products.

In the initial months (July–September), large order quantities were placed to build up stock levels in anticipation of future demand. The order placed in December 2025 was aimed at covering the raw material needs for January and February 2026.

Since there was no production in April (due to the shutdown period), demand dropped to zero. Overall, the planning effectively accounted for the 2 month lead time, avoiding both production disruptions and excessive stock accumulation.

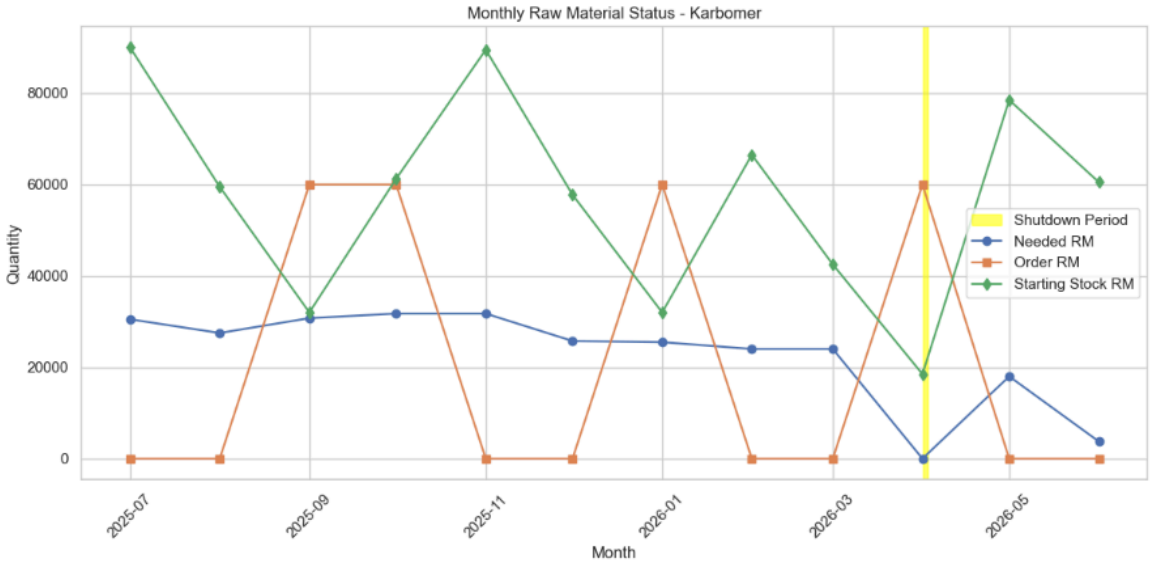

Figure 16 Monthly Raw Material Status of Karbomer

For the raw material Karbomer, production demand remained generally stable up until the shutdown. Orders were placed in a planned manner based on stock levels, and in some months, no orders were needed due to high stock availability. With the shutdown in April 2026, production demand dropped to zero, which is clearly reflected in the graph**.**

3.2 Literature Review

Accurate demand forecasting and its integration into planning systems are crucial for effective production planning. In their study, Grobler-Dębska and Andrzejczak (2022) proposed a practical approach to implementing a demand forecasting module within ERP systems, particularly in mass customization environments. The study focused on how forecasting techniques such as Moving Average, Weighted Moving Average, and Holt-Winters can be embedded directly into the ERP infrastructure. The authors demonstrated that when forecasting models are effectively integrated, production responsiveness and planning accuracy improve significantly. This is especially relevant in industries with fluctuating demand, such as pharmaceuticals. The study supports the use of classical time series models like Holt-Winters also applied in this project and shows their adaptability within corporate planning systems, bridging the gap between technical forecasting and real-world ERP application.

Another significant contribution comes from Bülbül and Kılıç (2024), who investigated how periodic updates in demand forecasts affect the performance of MRP systems. Through simulation-based experiments, the study revealed that frequent forecast revisions can have substantial effects on key planning outputs such as inventory levels, safety stock, and order timing. The findings emphasize that small changes in input forecasts may lead to large fluctuations in procurement and production activities. This insight directly aligns with the challenges addressed in the present project, where Excel-based static planning systems in the company were found to be inflexible against changing forecast data. In response, the development of a Python based interface that dynamically links forecast inputs to MPS and MRP outputs reflects the same principle explored in the study namely, that adaptive planning systems are essential for modern supply chain operations. Together, these studies contribute meaningfully to the literature by showing both the technical and operational value of integrated, responsive planning models.

3.3. Model Building Development of a Python Based Interface

*Figure SEQ Figure * ARABIC 17 MRP and MPS Interface*

This interface is designed to perform MPS and MRP calculations based on user inputs. The user begins by selecting the product, entering 12 month forecast data separated by commas, and specifying the starting stock level. Once the “Calculate” button is clicked, the backend developed in Python processes the inputs and generates results. The interface is visually aligned with Menarini’s corporate branding and provides a clean, user friendly layout. It eliminates the need for manual Excel operations by automating the entire planning workflow.

MPS

	Forecast	Starting Stock	Production Quantity	Stock on hand	Ending Stock	Stock Level (Month)
Ay 1	120155	125000	260000	4845	264845	2.05
Ay 2	127639	264845	280000	137206	417206	2.05
Ay 3	130243	417206	460000	286963	746963	2.02
Ay 4	277441	746963	240000	469522	709522	2.04
Ay 5	462915	709522	480000	246607	726607	2.03
Ay 6	232626	726607	240000	493981	733981	2.04
Ay 7	483138	733981	240000	250843	490843	2.06
Ay 8	235759	490843	100000	255084	355084	2.08
Ay 9	240863	355084	0	114221	114221	0.46
Ay 10	100145	114221	500000	14076	514076	2.02

MRP (Malzeme Planı)

Ethyl Alcohol

	Production Quantity	Needed Ethyl Alcohol	Starting Stock	Stock on hand	Order Quantity	Safety St
Ay 1	260000	130000	250000	120000	100000	185
Ay 2	280000	140000	220000	80000	100000	175
Ay 3	460000	230000	180000	0	300000	180
Ay 4	240000	120000	250000	130000	100000	180
Ay 5	480000	240000	230000	0	200000	120

*Figure SEQ Figure * ARABIC 19 MPS and MRP Overview Including Forecast, Production, and Raw Material Requirements*

Based on the inputs, this screen displays the calculated MPS and MRP results along with dynamic visualizations. The top section presents the MPS table, which includes details such as forecast, production quantity, stock on hand, and ending stock. Just below, a graph visualizes the relationship between forecast, production, and stock across 12 months, making it easier to interpret planning decisions. The lower section provides the MRP results for the relevant raw material, showing the monthly required quantity, available inventory, order quantity, and safety stock levels. With this structure, the interface allows users to efficiently evaluate both production planning and raw material needs in a single, integrated platform.

5.Conclusion

During this internship, I had the opportunity to closely examine production planning processes and apply my theoretical knowledge in a real business environment while developing a solution to a concrete problem. In particular, through my work on MPS (Master Production Scheduling) and MRP (Material Requirements Planning) calculations, I designed a tangible system using both Excel and Python.

As part of the project, I generated demand forecasts using historical sales data and various forecasting methods, and tested their accuracy through error analysis. I determined forecast data using the Holt-Winters method, which yielded the best results, and then carried out production and inventory planning based on these forecasts. In the existing system, calculations were done manually and tracked across multiple Excel sheets, which increased the risk of errors. To solve this problem, I integrated MPS and MRP into a single structure and also developed a user-friendly interface to automate the process.

Throughout my work, I paid special attention to incorporating various real-world factors into the model. For example, factors such as shutdown periods, the seasonality of demand forecasts, and lead times were included in the model to design a more realistic and applicable system. Therefore, a significant advantage of my work is that it is based not on fixed assumptions, but on a dynamic and flexible structure. However, the system can be further improved by testing it under different scenarios and expanding its scalability. Additionally, the visualization work done made inventory levels and production requirements more understandable.

From my own perspective, this process taught me how a real production planning cycle operates, the importance of data driven decision making, and how to analyze and improve a system from an industrial engineering point of view. By integrating my software knowledge with production management, I gained the ability to bring together different disciplines.

6.Bibliography

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Appendices

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